Linear Independent Component Analysis via Optimal Transport Metric as Contrast

Abstract

Independent Component Analysis (ICA) recovers statistically independent sources from observed linear mixtures. Classical algorithms—FastICA, JADE, and InfoMax—approximate non-Gaussianity via proxy contrast functions (negentropy approximations, fourth-order cumulants, parametric log-likelihoods) that fail systematically on heterogeneous source distributions. We propose OT-ICA, which quantifies marginal non-Gaussianity using the 1D squared Wasserstein distance (W_2^2) to a standard Gaussian, computed exactly by quantile sorting without density estimation. We prove W_2^2 satisfies a weighted mixture upper bound and a strict inequality for non-Gaussian sources, establishing it as a valid ICA contrast. Across $d \in \{10, 20, 30\}$ and five mixture regimes ($N = 10{,}000$, 10 trials), OT-ICA achieves lower Amari error than FastICA, JADE, InfoMax, and Picard on all continuous and heterogeneous distributions. Two applications are presented: ocular artifact isolation in frontal EEG recordings, and structural shock identification in Vector Error Correction Models for financial price discovery.

1 INTRODUCTION

Blind source separation, specifically formulated as Independent Component Analysis (ICA) [Jutten and Herault, 1985, Comon, 1994], is a fundamental problem in unsupervised probabilistic modeling. The objective is to recover a set of hidden, mutually independent source signals $\mathbf{S}$ from an observed linear mixture $\mathbf{X}$. The theoretical foundation of this recovery relies on strict identifiability conditions.

Theorem 1 (The Linear ICA Model and Identifiability [Comon, 1994])**.** *Let the observed data be denoted by a random vector $\mathbf{X} \in \mathbb{R}^d$, and the latent sources by $\mathbf{S} \in \mathbb{R}^d$. The linear ICA model assumes:*

1. ***Linear Mixing:*** $\mathbf{X} = \mathbf{AS}$*, where $\mathbf{A} \in \mathbb{R}^{d\times d}$ is an unknown, invertible mixing matrix.*
2. ***Independence:*** *The components $s_1, \dots, s_d$ of $\mathbf{S}$ are mutually statistically independent.*
3. ***Non-Gaussianity:*** *At most one of the source components s_i follows a Gaussian distribution.*

Under these conditions, the mixing matrix $\mathbf{A}$ can be uniquely identified from the observations $\mathbf{X}$, up to a permutation and scaling of its columns.

Because of the Central Limit Theorem, mixtures of independent non-Gaussian variables are mathematically more Gaussian than the constituent sources. Standard ICA isolates these sources by navigating the space of orthogonal rotation matrices to maximize the non-Gaussianity of the projected data [Hyvärinen et al., 2001]. This search for maximum non-Gaussianity is grounded in information geometry.

Theorem 2 (Equivalence of Independence and Non-Gaussianity [Cardoso, 2022])**.** *Let Y be a zero-mean random vector with an identity covariance matrix (whitened). The mutual information $\mathscr{I}(Y)$ is directly determined by the joint non-Gaussianity $G(Y)$ and the sum of marginal non-Gaussianities $\sum_i G(Y_i)$ via the strict relationship:*

$$\mathscr{I}(Y) = G(Y) - \sum_i G(Y_i) \tag{1}$$

Because joint non-Gaussianity $G(Y)$ is invariant under orthogonal rotation, minimizing mutual information (extracting independent components) is mathematically equivalent to maximizing the sum of marginal non-Gaussianities.

(A comprehensive proof of this theorem using Kullback-Leibler Pythagorean identities on the Product and Gaussian manifolds is provided in Appendix A.)

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However, evaluating the full marginal density is computationally expensive. Classical algorithms—FastICA [Hyvärinen et al., 2001, Amari et al., 1996], InfoMax [Bell and Sejnowski, 1995], JADE [Cardoso and Souloumiac, 1993], and Picard [Ablin et al., 2018]—replace it with surrogate contrast functions (*logcosh* approximations, log-likelihood surrogates, fourth-order cumulants, and adaptive nonlinearities). These proxies fail systematically on heterogeneous source distributions (detailed in Appendix E).

Prior work applies 1D W_2 to normality testing [del Barrio et al., 1999] and marginal distribution comparison [Peyré and Cuturi, 2019], but does not formalize W_2^2-to-Gaussian as an ICA contrast function. We propose OT-ICA. In optimal transport theory, the Monge problem [Monge, 1781] and its Kantorovich relaxation [Kantorovich, 1942] define the general Wasserstein distance W_p [Villani, 2003] between a source measure μ and a target ν over the space of optimal probabilistic couplings $\Pi(\mu, \nu)$ as:

$$W_p(\mu,\nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int \|x-y\|^p \, d\pi(x,y) \right)^{1/p} \quad (2)$$

While calculating this coupling is intractable in high dimensions, in the 1D setting data can be unambiguously sorted. By Brenier's Theorem [Brenier, 1991] and Caffarelli's regularity theory [Caffarelli, 1992], the 1D transport map is monotonic. Consequently, the squared Wasserstein distance ($p = 2$) analytically collapses to the integrated squared difference between the inverse cumulative distribution functions (quantile functions):

$$W_2^2(\mu,\nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^2 \, dt \quad (3)$$

By restricting to 1D projections, OT-ICA computes a proxy-free metric exactly via quantile sorting, without density estimation.

2 THE OT-ICA FRAMEWORK

Once observed data $\mathbf{X}$ is centered and whitened to yield unit-variance and uncorrelated components $\widetilde{\mathbf{X}}$, the search for independent components is strictly constrained to the Orthogonal Group $\mathcal{S}$ [Cardoso, 2022]. The OT-ICA objective is to find the orthogonal projection $\mathbf{b}$ that maximizes the distance to the standard normal distribution $\Gamma \equiv \mathcal{N}(0,1)$:

$$\hat{\mathbf{b}} = \arg\max_{\mathbf{b} \in \mathcal{S}} W_2^2\big(\mathbb{P}_{\mathbf{b}^\top \widetilde{\mathbf{X}}}, \Gamma\big). \quad (4)$$

This maximisation is justified by the following bounds on W_2^2 for linear mixtures.

Theorem 3 (W_2 Upper Bound)**.** *Let $\mathbf{Z} = (Z_1, \ldots, Z_d)^\top$ be a vector of mutually independent latent sources, and let $\mathbf{b}$ be a weight vector on the unit sphere ($\sum_{i=1}^d b_i^2 = 1$). The squared Wasserstein distance between the linear mixture $\mathbf{b} \cdot \mathbf{Z}$ and the standard normal distribution Γ is bounded by the weighted sum of the source distances to Γ:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \le \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (5)$$

Theorem 4 (Strict W_2 Inequality for Non-Gaussian Sources)**.** *Assuming at most one source Z_i is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3 is a strict inequality for any valid mixture where multiple weights b_i are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (6)$$

The global maximum is therefore attained when $\mathbf{b}$ aligns with a single source direction, isolating one independent component. *(Full proofs via comonotonicity contradictions under Caffarelli's regularity theory are in Appendix B.)*

2.1 METHODOLOGICAL ENHANCEMENTS

Optimization of W_2^2 on the Orthogonal Group requires three enhancements: (i) **Analytical Gaussian Targets** [Fournier and Guillin, 2015] to eliminate target-sampling noise; (ii) **Riemannian Gradient Retraction** onto the Orthogonal Group [Absil et al., 2008]; and (iii) **Gaussian Dithering** to smooth discrete-CDF plateaus [Schuchman, 1964]. Full formulations and pseudocode are in Appendix C.

3 OT-ICA METHODOLOGY VALIDATION

We evaluate separation quality via the Amari Performance Index (E) [Amari et al., 1996], which measures the deviation of the gain matrix $\mathbf{P} = \mathbf{B}_{\text{est}}\mathbf{A}$ from a generalized permutation matrix. $E < 0.1$ indicates near-perfect separation; $E \ge 0.5$ indicates failure (formal definition in Appendix D).

Figure 1 shows that W_2^2 drives the Amari error to near zero on four standard continuous source types, confirming convergence of the Riemannian solver with analytical Gaussian targets.

4 EMPIRICAL ROBUSTNESS ON HYBRID MIXTURES

We benchmark OT-ICA, FastICA [Hyvärinen et al., 2001], JADE [Cardoso and Souloumiac, 1993], InfoMax [Bell and Sejnowski, 1995], and Picard [Ablin et al., 2018] across five mixture regimes at $d \in \{10, 20, 30\}$, $N = 10{,}000$, 10 trials per condition.

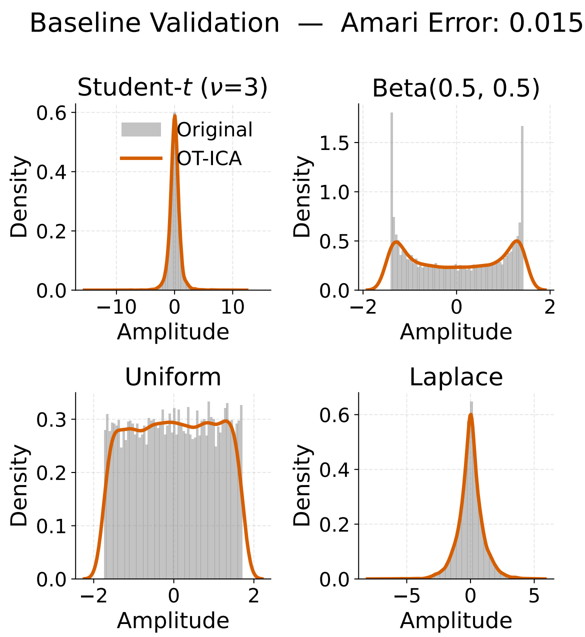

Figure 1: Baseline validation of OT-ICA convergence on standard continuous mixtures. The squared Wasserstein distance (W_2^2) acts as a smooth contrast, driving the Amari error toward zero.

Figure 2 shows three structurally distinct proxy failures at $d = 30$, each attributable to a different mechanism. **FastICA** fails via (i) *zero negentropy*: sources satisfying $\mathbb{E}[G(x)] = \mathbb{E}[G(\nu)]$ produce no contrast gradient; and (ii) *vanishing curvature*: the Newton-update denominator $\mathbb{E}[sg(s) - g'(s)] \to 0$, making the fixed-point step undefined (Appendix E). **Picard** [Ablin et al., 2018] uses the same tanh nonlinearity as FastICA but with a provably converging orthogonal solver; its performance tracks FastICA on heterogeneous mixtures, confirming that the failure originates in the contrast function rather than the optimizer. **JADE** requires $\mathcal{O}(d^4)$ samples for reliable cumulant estimation [Cardoso and Souloumiac, 1993]—≈810,000 at $d = 30$ versus $N = 10{,}000$ available—so its joint diagonalisation is noise-dominated. **InfoMax**'s parametric log-likelihood [Bell and Sejnowski, 1995] is misspecified on distributions mixing super- and sub-Gaussian sources simultaneously. OT-ICA achieves the lowest Amari error on all continuous and heterogeneous regimes by evaluating W_2^2 directly without parametric assumptions.

The *Discrete Only* failure is distinct: a *contrast* failure for FastICA but an *optimization* problem for OT-ICA. Table 2 (Appendix E) shows W_2^2 provides $\gg 10\times$ the non-Gaussianity resolution of logcosh—standard Poisson ($\lambda = 3$) has near-zero logcosh negentropy but a clear W_2^2 signal of 0.05. The step-function CDFs of discrete data create gradient plateaus that stall the solver despite the contrast being present. (Scaling in Appendix F.)

5 INTERDISCIPLINARY APPLICATIONS

5.1 STRUCTURAL IDENTIFICATION IN PRICE DISCOVERY

In fragmented financial markets, cointegrated asset prices share an unobserved common random walk. The Information Share (IS) [Hasbrouck, 1995, Baillie et al., 2002] of each venue quantifies its contribution, but requires decomposing the reduced-form residuals u_t of a Vector Error Correction Model (VECM) [Engle and Granger, 1987, Johansen, 1991] into independent structural shocks ϵ_t via $u_t = \mathbf{B}\epsilon_t$.

The covariance constraint $\mathbf{\Omega} = \mathbf{B}\mathbf{B}^\top$ leaves $\mathbf{B}$ unidentified up to an orthogonal rotation. Standard Cholesky decomposition fixes this rotation by imposing a recursive market ordering, producing IS estimates that change with the ordering choice.

Factoring $\mathbf{B} = \mathbf{SC}$ (whitening $\times$ orthogonal rotation) reduces identification to finding $\mathbf{C}$ [Zema and Cordoni, 2025]. OT-ICA recovers $\mathbf{C}$ from the heavy-tailed non-Gaussianity of financial innovations without imposing a recursive structure.

Market / Source	True IS	Estimated IS (Mean)	Std. Dev.
Market 1	0.1200	0.1212	0.0161
Market 2	0.2400	0.2399	0.0245
Market 3	0.6400	0.6389	0.0260

Table 1: Simulation Results for Information Share Recovery across 500 Monte Carlo runs. The estimated IS tightly matches the true IS without imposing arbitrary Cholesky recursive structures.

Table 1 shows mean IS estimates within 0.2 percentage points of the true values across 500 Monte Carlo runs, with standard deviations below 0.03.

5.2 EEG ARTIFACT REMOVAL

The skull acts as a linear volume conductor [Hyvärinen et al., 2001]: ocular blink artifacts mix linearly with neural signals before reaching scalp electrodes. Blink artifacts are impulsive and super-Gaussian; OT-ICA identifies them by maximising W_2^2 without assuming a parametric density model.

Figure 3 shows OT-ICA concentrating the blink artifact into one component with substantially higher excess kurtosis than the remaining four. Zeroing that component and reconstructing the signal produces a measurable RMS reduction in the blink window (quantified in the panel title), with no manual tuning of a density model.

6 CONCLUSION

OT-ICA formulates ICA as a 1D optimal transport problem, replacing proxy contrast functions with the exact W_2^2 metric. It avoids FastICA's gradient degeneracy, JADE's fourth-order sample complexity, and InfoMax's parametric model assumptions; the Picard comparison confirms that faster optimisation alone does not close the gap. Across $d \in \{10, 20, 30\}$ and five mixture regimes, OT-ICA achieves lower Amari error on all continuous and heterogeneous distributions. Applications to VECM price discovery and EEG artifact removal extend the method beyond standard benchmarks. Future work will extend OT-ICA to causal discovery (LiNGAM [Shimizu et al., 2006], CauCA [Liang et al., 2023]).

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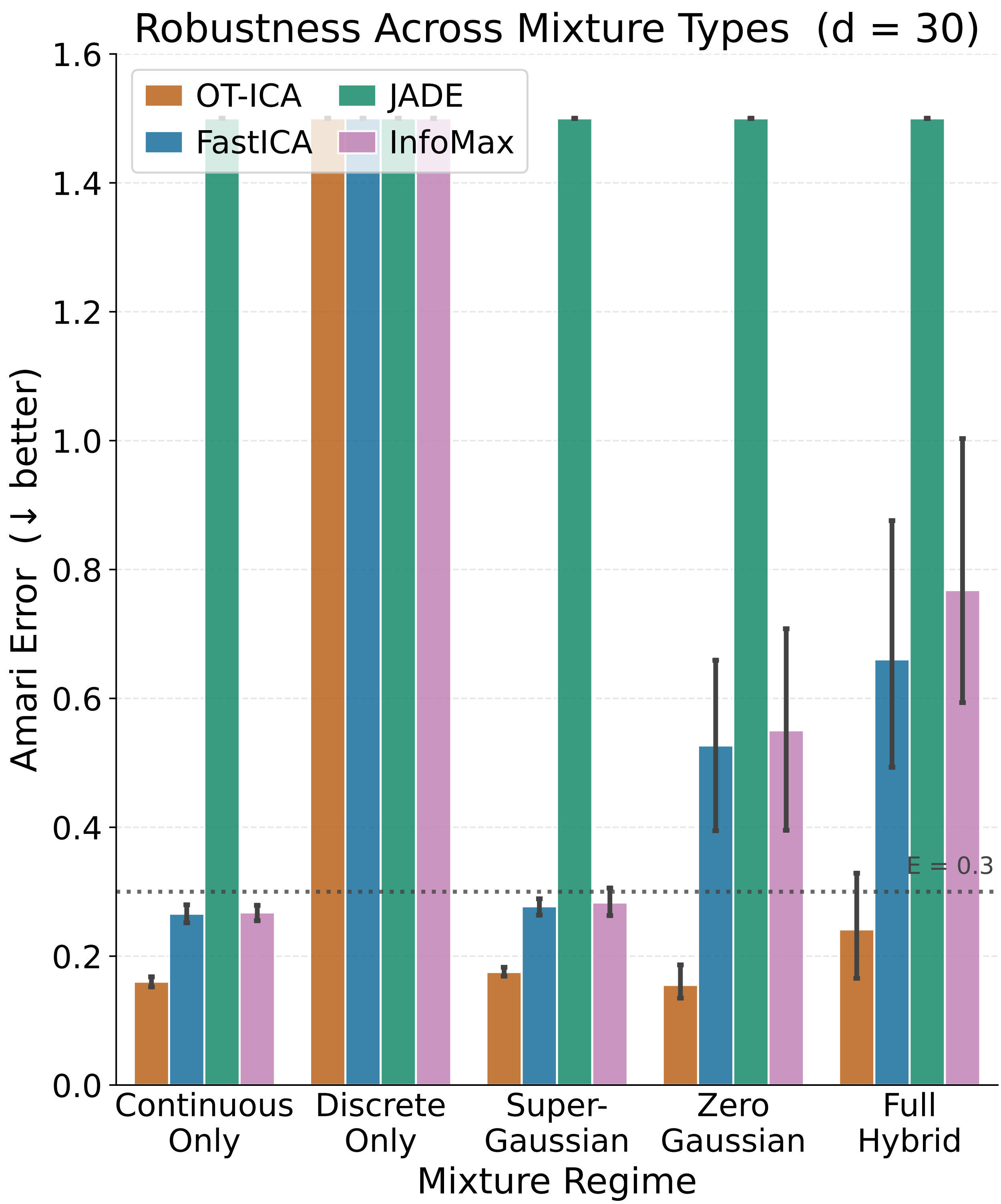

Figure 2: Amari error (↓ better) across five mixture regimes and $d \in \{10, 20, 30\}$ ($N = 10{,}000$, 10 trials, 95% CI). OT-ICA achieves the lowest error on all continuous and heterogeneous regimes at every dimension. At $d = 30$ the proxy-based methods separate into three failure modes (Section 4). On *Discrete Only* all methods degrade—a gradient plateau in the W_2^2 optimization surface, not a contrast failure (Appendix E).

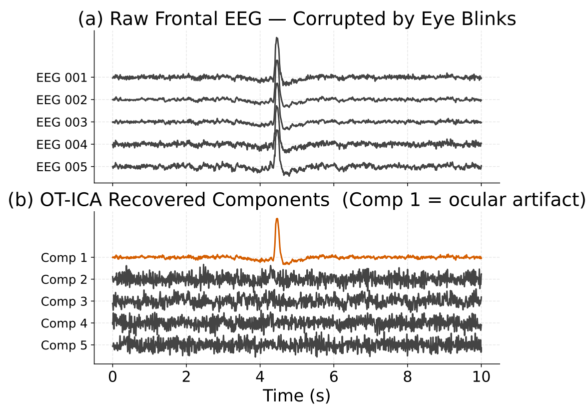

Figure 3: OT-ICA on five frontal EEG channels (MNE sample dataset). Panel (b) shows the isolated blink component (orange) with excess kurtosis κ reported in the panel title; the remaining components have mean κ near zero. Panel (c) shows the reconstructed signal after zeroing the artifact component, with the RMS reduction in the ± 250 ms blink window reported in the panel title.

Supplementary Material for:
Linear Independent Component Analysis via Optimal Transport Metric as Contrast

A INFORMATION GEOMETRY OF ICA

Information geometry provides a framework for quantifying the relationship between independence and non-Gaussianity by analyzing probability distributions as points on geometric manifolds. Following the perspective introduced by Cardoso [2022], ICA can be understood as orthogonally projecting the empirical distribution P_Y onto two distinct subspaces: the Product manifold $\mathcal{P}$ (containing distributions where marginals are strictly independent) and the Gaussian manifold $\mathcal{G}$ (encompassing all multivariate Gaussians).

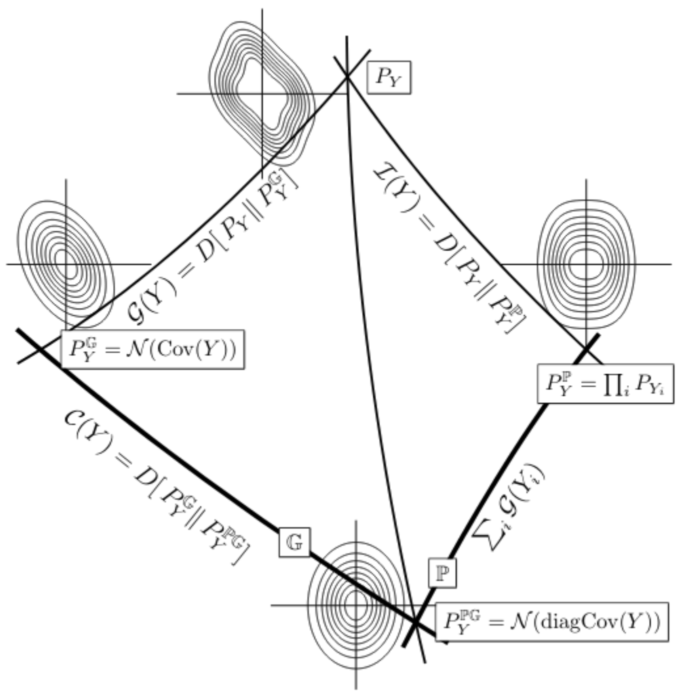

Figure 4: The Pythagorean geometry of Independent Component Analysis. Figure from Cardoso [2022].

The Mutual Information $\mathscr{I}(Y)$ measures the distance from P_Y to its closest independent product distribution P_Y^P. A Pythagorean identity exists on the Product manifold for any candidate product distribution $Q = \prod_i q_i$:

$$
\begin{aligned}
D_{\mathrm{KL}}\left(P_Y \| Q\right) &= \int P_Y(y) \log\left(\frac{P_Y(y)}{P_Y^P(y)} \frac{P_Y^P(y)}{Q(y)}\right) dy \\
&= \mathscr{I}(Y) + \sum_i D_{\mathrm{KL}}\left(P_{Y_i} \| q_i\right).
\end{aligned} \tag{7}
$$

A similar Pythagorean identity holds for the Gaussian manifold $\mathcal{G}$. Let $\Phi_Y = \mathcal{N}(0, \operatorname{Cov} Y)$ be the best Gaussian fit (sharing the exact mean and covariance of P_Y), and let $\Phi = \mathcal{N}(0, \Sigma)$ be an arbitrary Gaussian target. The divergence decomposes as:

$$
\begin{aligned}
D_{\mathrm{KL}}\big(P_Y \| \Phi\big) &= \int P_Y(y) \log\left(\frac{P_Y(y)}{\Phi_Y(y)} \frac{\Phi_Y(y)}{\Phi(y)}\right) dy \\
&= \int P_Y(y) \log \frac{P_Y(y)}{\Phi_Y(y)} dy + \int P_Y(y) \log \frac{\Phi_Y(y)}{\Phi(y)} dy \\
&= D_{\mathrm{KL}}\big(P_Y \| \Phi_Y\big) + \mathbb{E}_{P_Y}\left[\log \Phi_Y(y) - \log \Phi(y)\right].
\end{aligned} \tag{8}
$$

Because the log-density of a zero-mean Gaussian is a quadratic form, its expectation strictly depends only on the second moments (the covariance matrix) of the variable. Since the true distribution P_Y and its Gaussian fit Φ_Y share the exact same covariance matrix by definition, evaluating the expectation of this quadratic form under P_Y is mathematically identical to evaluating it under Φ_Y:

$$\mathbb{E}_{P_Y}\left[\log\frac{\Phi_Y(y)}{\Phi(y)}\right] = \mathbb{E}_{\Phi_Y}\left[\log\frac{\Phi_Y(y)}{\Phi(y)}\right] = D_{\mathrm{KL}}\big(\Phi_Y \|\Phi\big). \tag{9}$$

Substituting this equivalence back establishes the Pythagorean identity on the Gaussian manifold:

$$D_{\mathrm{KL}}\big(P_Y\|\mathcal{N}(\Sigma)\big) = D_{\mathrm{KL}}\big(P_Y\|\mathcal{N}(\operatorname{Cov}Y)\big) + D_{\mathrm{KL}}\big(\mathcal{N}(\operatorname{Cov}Y)\|\mathcal{N}(\Sigma)\big). \tag{10}$$

We define the joint non-Gaussianity as $G(Y) = D_{\mathrm{KL}}(P_Y \,\|\, \mathcal{N}(\operatorname{Cov}Y))$, and correlation as $C(Y) = D_{\mathrm{KL}}(\mathcal{N}(\operatorname{Cov}Y)\,\|\,\mathcal{N}(\operatorname{diag}\operatorname{Cov}Y))$. By evaluating the KL divergence from P_Y to a fully factorized Gaussian target $\mathcal{N}(\operatorname{diag}\operatorname{Cov}Y)$ via both the Product path and Gaussian path, we yield Cardoso's unified theorem [Cardoso, 2022]:

$$\mathscr{I}(Y) + \sum_i G(Y_i) = G(Y) + C(Y). \tag{11}$$

Crucially, once the data is whitened ($\operatorname{Cov}(Y) = \mathbf{I}$), correlation drops strictly to zero ($C(Y) = 0$). The relationship rearranges strictly as:

$$\mathscr{I}(Y) = G(Y) - \sum_i G(Y_i). \tag{12}$$

Because joint non-Gaussianity $G(Y)$ is invariant under orthogonal rotation, minimizing mutual information (achieving independence) mathematically equates exactly to maximizing the sum of marginal non-Gaussianities $\sum G(Y_i)$.

B MATHEMATICAL PROOFS FOR OT-ICA BOUNDS

B.1 PROOF OF THEOREM 3: W_2 UPPER BOUND

Theorem 5 (W_2 Upper Bound)**.** *Let $\mathbf{Z} = (Z_1, \ldots, Z_d)^\top$ be a vector of mutually independent latent sources, and let $\mathbf{b}$ be a weight vector on the unit sphere ($\sum_{i=1}^d b_i^2 = 1$). The squared Wasserstein distance between the linear mixture $\mathbf{b}\cdot\mathbf{Z}$ and the standard normal distribution Γ is bounded by the weighted sum of the source distances to Γ:*

$$W_2^2(\mathbf{b}\cdot\mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \tag{13}$$

Proof. We construct a specific joint probability distribution, or coupling, using a common source of randomness. Let $\mathbf{N} = (N_1, \ldots, N_d)^\top$ be a vector of d independent standard normal variables, $\mathbf{N} \sim \mathcal{N}(0, \mathbf{I}_d)$, assumed to be statistically independent of the sources $\mathbf{Z}$. For each independent latent source Z_i, there exists an optimal transport map T_i that pushes forward the standard normal distribution to the source distribution, denoted as $(T_i)_\#\Gamma = Z_i$.

We construct a random variable X representing the mixture:

$$X = \sum_{i=1}^d b_i T_i(N_i). \tag{14}$$

To compare this mixture to a Gaussian distribution, we construct a target variable Y using the same underlying noise vector $\mathbf{N}$:

$$Y = \sum_{i=1}^d b_i N_i. \tag{15}$$

Given that the weights sum to unity, the resulting variable Y follows the standard normal distribution, $Y \sim \Gamma$. The pair (X, Y) creates a valid coupling. Because W_2^2 is the infimum over all valid couplings, the optimal distance is less than or equal to the cost of our constructed coupling:

$$W_2^2(\mathbf{b}\cdot\mathbf{Z}, \Gamma) \leq \mathbb{E}\left[|X - Y|^2\right] = \mathbb{E}\left[\left(\sum_{i=1}^d b_i\big(T_i(N_i) - N_i\big)\right)^2\right]. \tag{16}$$

Squaring the summation produces diagonal terms ($i = j$) and cross terms ($i \neq j$):

$$|X - Y|^2 = \sum_{i=1}^{d} b_i^2 \big(T_i(N_i) - N_i\big)^2 + \sum_{i \neq j} b_i b_j \big(T_i(N_i) - N_i\big)\big(T_j(N_j) - N_j\big). \tag{17}$$

Because the underlying noise variables N_i and N_j are independent, and the functions evaluate to a mean of zero, when we take the expectation, all cross terms vanish. We are left with the expectation of the diagonal terms:

$$\mathbb{E}\left[|X - Y|^2\right] = \sum_{i=1}^{d} b_i^2 \mathbb{E}\left[\big(T_i(N_i) - N_i\big)^2\right]. \tag{18}$$

Recognizing that $\mathbb{E}[(T_i(N_i) - N_i)^2]$ is the definition of the squared Wasserstein distance between the individual source Z_i and Γ, we arrive at the bound:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^{d} b_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \tag{19}$$

B.2 PROOF OF THEOREM 4: STRICT INEQUALITY

Theorem 6 (Strict W_2 Inequality for Non-Gaussian Sources)**.** *Assuming at most one source Z_i is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3 is a strict inequality for any valid mixture where multiple weights b_i are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^{d} b_i^2 W_2^2(Z_i, \Gamma). \tag{20}$$

Proof. In a one-dimensional setting, the W_2 distance equals the expected squared difference of a specific coupling if and only if the random variables are comonotonic. By Brenier's Theorem [Brenier, 1991], this requires the gradients to be parallel at every point in the probability space:

$$\nabla X(\mathbf{N}) = \lambda(\mathbf{N}) \nabla Y(\mathbf{N}). \tag{21}$$

Assuming the source distributions possess smooth and strictly positive densities, Caffarelli's regularity theory [Caffarelli, 1992] guarantees that the optimal transport maps T_i are continuously differentiable.

We differentiate both sides to compute their Hessian matrices. On the LHS, the cross-derivatives $\frac{\partial^2 X}{\partial N_i \partial N_j}$ are zero, yielding a diagonal matrix of $b_i T_i''(N_i)$. On the RHS, we differentiate the product $\lambda(\mathbf{N})\mathbf{b}$, yielding a Rank-1 outer product matrix:

$$\underbrace{\begin{bmatrix} b_1 T_1''(N_1) & 0 & \cdots & 0 \\ 0 & b_2 T_2''(N_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_d T_d''(N_d) \end{bmatrix}}_{\text{Hessian of } X \text{ (Diagonal)}} = \underbrace{\begin{bmatrix} b_1 \frac{\partial \lambda}{\partial N_1} & \cdots & b_1 \frac{\partial \lambda}{\partial N_d} \\ \vdots & \ddots & \vdots \\ b_d \frac{\partial \lambda}{\partial N_1} & \cdots & b_d \frac{\partial \lambda}{\partial N_d} \end{bmatrix}}_{\text{Outer Product } \mathbf{b}(\nabla \lambda)^\top \text{ (Rank-1)}} \tag{22}$$

Examining any off-diagonal entry ($i \neq j$), the LHS dictates it must be zero, establishing $b_i \frac{\partial \lambda}{\partial N_j} = 0$. Because we assume a mixture where multiple weights are non-zero ($b_i \neq 0$), it follows that the gradient of the scaling function is zero ($\nabla \lambda = \mathbf{0}$). Consequently, $\lambda(\mathbf{N})$ is a global scalar constant, λ.

Equating the diagonal elements implies $T_i'(N_i) = \lambda$. Integrating this indicates the optimal transport maps are linear functions of the form:

$$T_i(x) = \lambda x + c. \tag{23}$$

Applying a purely linear transformation to Gaussian noise N_i produces another Gaussian distribution. The identifiability assumption of ICA dictates that the original latent sources Z_i are non-Gaussian. Therefore, the linear map requirement contradicts the non-Gaussianity of the sources. Because the condition for equality cannot be met, the relationship is a strict inequality. $\blacksquare$

B.3 PROOF OF INTRACTABILITY OF NEWTON UPDATES FOR EMPIRICAL OT-ICA

Theorem 7 (Intractability of Newton Updates)**.** *For an empirical distribution mapped to a continuous target via sorting, computing a closed-form Hessian* $\mathbf{H} = \nabla^2_{\mathbf{b}} W_2^2$ *is analytically intractable without introducing external density estimation mechanisms. The Map Derivative* $\nabla_{\mathbf{b}} T_{\mathbf{b}}$ *strictly requires the continuous Probability Density Function (PDF) evaluated at the quantile boundaries.*

Proof. To implement a fixed-point Newton step, the Hessian matrix of the squared Wasserstein distance, $\mathbf{H} = \nabla^2_{\mathbf{b}} W_2^2$, is required. Let the 1D projection of the data be $y = \mathbf{b}^\top \mathbf{X}$, and let $T_{\mathbf{b}}$ be the optimal transport map pushing this empirical projection to the target Gaussian distribution.

We compute the first differential (the gradient) with respect to $\mathbf{b}$. Utilizing the envelope theorem for optimal transport, the variation of the optimal map evaluates to zero within the cost function:

$$\mathbf{g} = \nabla_{\mathbf{b}} \left(\frac{1}{2} W_2^2 \right) = \mathbb{E}\Big[\mathbf{X}\big(\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\big)\Big]. \tag{24}$$

To compute the Hessian $\mathbf{H}$, we must differentiate the gradient vector $\mathbf{g}$ with respect to $\mathbf{b}$:

$$\mathbf{H} = \nabla_{\mathbf{b}} \mathbf{g} = \mathbb{E}\Big[\mathbf{X}\mathbf{X}^\top - \mathbf{X} \nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})]^\top\Big]. \tag{25}$$

Applying the total derivative using the chain rule yields two distinct components:

$$\nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})] = \underbrace{T'_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})\mathbf{X}}_{\text{Argument Derivative}} + \underbrace{(\nabla_{\mathbf{b}} T_{\mathbf{b}})(\mathbf{b}^\top \mathbf{X})}_{\text{Map Derivative}}. \tag{26}$$

By Caffarelli's regularity theory [Caffarelli, 1992], the transport map $T_{\mathbf{b}}$ is continuously differentiable, guaranteeing the existence of the Argument Derivative. The fundamental limitation lies within the Map Derivative.

In the 1D optimal transport setting, the transport map relies on the empirical cumulative distribution function, $F_{\mathbf{b}}$, of the projected data y. Expanding the Map Derivative yields:

$$(\nabla_{\mathbf{b}} T_{\mathbf{b}})(y) = \nabla_{\mathbf{b}} \big[\Phi^{-1}(F_{\mathbf{b}}(y))\big] = \frac{1}{\phi(\Phi^{-1}(F_{\mathbf{b}}(y)))} \nabla_{\mathbf{b}} F_{\mathbf{b}}(y). \tag{27}$$

The term $\nabla_{\mathbf{b}} F_{\mathbf{b}}(y)$ represents the rate of change of the cumulative probability mass as the projection plane $\mathbf{b}$ is rotated. The derivative of a cumulative distribution function's boundary strictly requires the underlying continuous probability density function (PDF) evaluated at that specific boundary. For empirical distributions consisting of discrete sampled points, this density does not natively exist, rendering the closed-form calculation of the Hessian analytically intractable. ■

C ALGORITHMIC ENHANCEMENTS AND METHODOLOGY

C.1 WHY W_2^2 INSTEAD OF W_1?

While the W_1 distance is robust to outliers, its reliance on an absolute L_1 penalty introduces severe geometric challenges on finite samples.

As shown in Figure 5, the W_1 cost lacks the convexity needed to smooth out empirical noise. Its derivative fluctuates frequently, causing first-order gradient ascent solvers to stall in local non-smooth traps. The squared penalty in the W_2^2 metric acts as a structural smoother, penalizing local distortions and providing a continuous, decelerating gradient profile that allows for reliable convergence.

C.2 ANALYTICAL GAUSSIAN TARGETS

Discrete approximation of the target Gaussian distribution introduces approximation noise. To eliminate this noise, we replace point-sampled targets with an analytical formulation, computing the expected value of a standard normal variable within each discrete quantile bin.

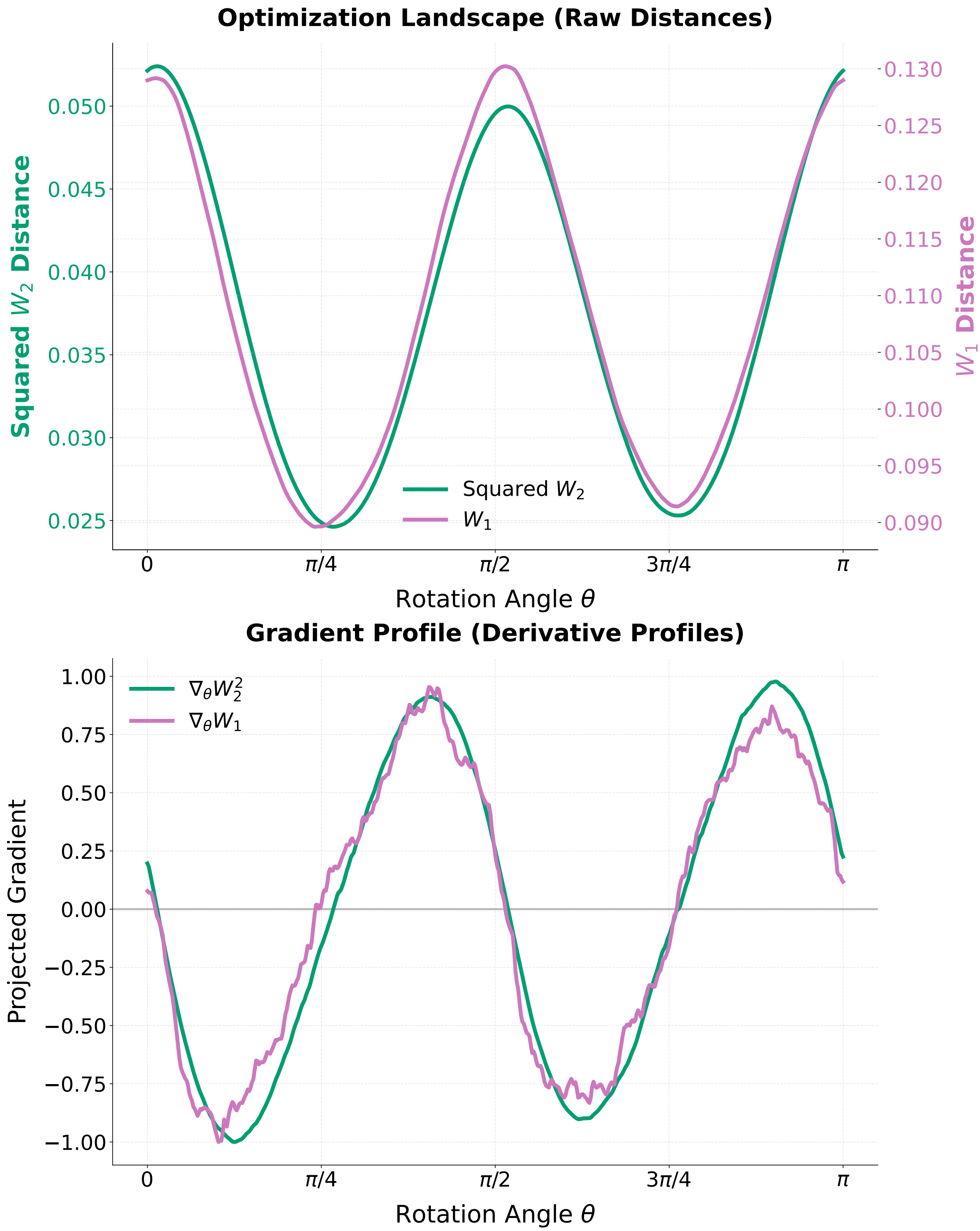

Figure 5: Gradient Landscapes of W_1 and W_2^2 across continuous rotation angles θ. The W_2^2 gradient smoothly decelerates toward zero, whereas the W_1 gradient exhibits jagged volatility on finite empirical samples.

Definition 1 (Analytical Gaussian Target)**.** *Let the uniform probability bin edges for N samples be defined as $p_i = \frac{i}{N}$ for $i = 0, \ldots, N$, with corresponding Gaussian domain boundaries $z_i = \Phi^{-1}(p_i)$, where Φ is the standard normal CDF. The analytical target value T_i for the i-th sorted sample is:*

$$T_i = N \int_{z_{i-1}}^{z_i} x\phi(x)\, dx = N\big(\phi(z_{i-1}) - \phi(z_i)\big). \tag{28}$$

C.3 RIEMANNIAN GRADIENT AND RETRACTION

Because the input data is whitened, the unmixing matrix $\mathbf{B}$ must remain in the Orthogonal Group $\mathcal{S}$, satisfying $\mathbf{B}\mathbf{B}^\top = \mathbf{I}$. Computing the standard Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{B}} W_2^2$ points into the unconstrained ambient space.

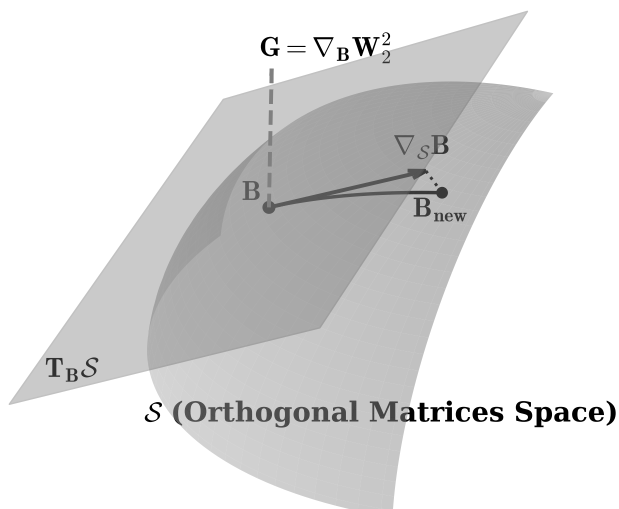

Figure 6: Geometric sketch of optimization on the Orthogonal Group $\mathcal{S}$. The Euclidean gradient $\mathbf{G}$ is projected onto the tangent space to yield the Riemannian gradient $\nabla_{\mathcal{S}}\mathbf{B}$. A retraction maps the estimate back to the orthogonal surface.

Definition 2 (Riemannian Gradient on the Orthogonal Group)**.** *Given the Euclidean gradient $\mathbf{G}$, the Riemannian gradient resulting from the projection onto the tangent space $T_{\mathbf{B}}\mathcal{S}$ is given by [Absil et al., 2008]:*

$$\nabla_{\mathcal{S}}\mathbf{B} = \mathbf{G} - \frac{1}{2}\big(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top\big)\mathbf{B}. \tag{29}$$

Definition 3 (Symmetric Decorrelation Retraction)**.** *To map the matrix back to the orthogonal surface after a tangent step ($\mathbf{B}_{step}$), we compute the overlap covariance $\mathbf{C} = \mathbf{B}_{step}\mathbf{B}_{step}^\top$ and apply the inverse square root:*

$$\mathbf{B}_{new} = \mathbf{C}^{-1/2}\mathbf{B}_{step}. \tag{30}$$

C.4 CONTINUOUS SMOOTHING VIA GAUSSIAN DITHERING

Applying continuous optimal transport directly to discrete mixtures introduces non-smooth step-functions. Adapted from signal processing [Schuchman, 1964], we inject a variance of continuous, zero-mean Gaussian noise ($\sigma^2_{\text{dither}} = 0.01$) into the 1D projected components prior to sorting. This decorrelates deterministic quantization errors and convolves the discrete empirical distribution with a Gaussian kernel.

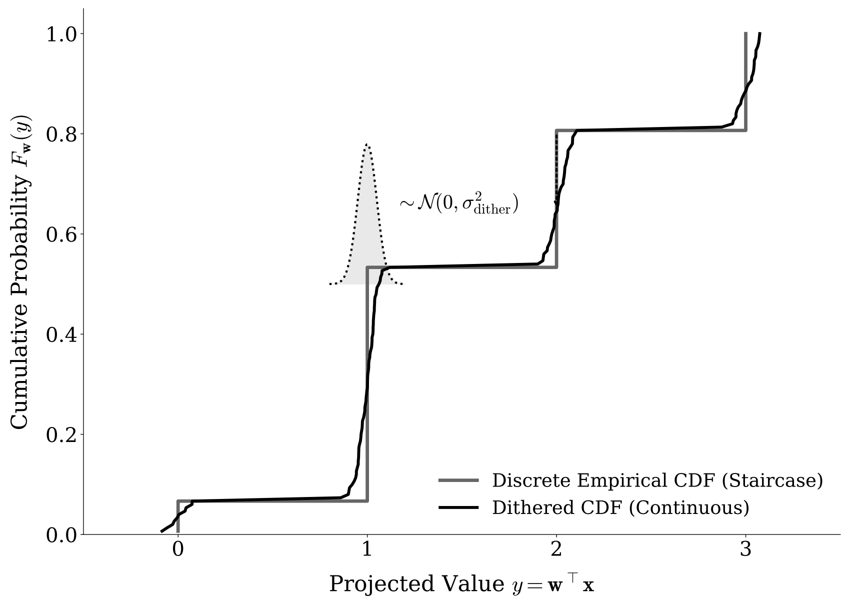

Figure 7: The non-differentiable staircase of a discrete empirical CDF (gray) is convolved with a narrow Gaussian kernel via dithering, yielding a continuous, differentiable curve (black).

C.5 TOTAL COMPLEXITY AND STATISTICAL EFFICIENCY

For a dataset of dimension d with N samples, K random restarts (batched into a tensor $\mathbf{B}_{\text{batch}}$), and T optimization iterations, the per-iteration complexity is:

$$\mathcal{O}(K \cdot (d \cdot N \log N + d^3)) \tag{31}$$

The $N \log N$ term represents the sorting of projections, while the d^3 term represents the cost of the symmetric decorrelation used for retraction.

C.6 THE OT-ICA ALGORITHM

D EXPERIMENTAL METHODOLOGY AND METRICS

D.1 THE GENERALIZED PERMUTATION MATRIX AND AMARI INDEX

To empirically evaluate the success of source separation, we quantify the distance between the estimated unmixing matrix $\mathbf{B}_{\text{est}}$ and the true mixing matrix $\mathbf{A}$. Because ICA is subject to scaling and permutation ambiguities, the target recovery is a generalized permutation matrix. Let $\mathbf{P} = \mathbf{B}_{\text{est}}\mathbf{A}$ be the global transfer matrix. For a $d \times d$ matrix $\mathbf{P}$ with elements p_{ij}, the Amari Performance Index [Amari et al., 1996] measures the structural divergence:

$$E(\mathbf{P}) = \frac{1}{2d}\sum_{i=1}^{d}\left(\sum_{j=1}^{d}\frac{|p_{ij}|}{\max_k |p_{ik}|} - 1\right) + \frac{1}{2d}\sum_{j=1}^{d}\left(\sum_{i=1}^{d}\frac{|p_{ij}|}{\max_k |p_{kj}|} - 1\right). \tag{32}$$

D.2 COMPUTATIONAL RESOURCE REGIMES

Because OT-ICA utilizes stochastic gradient descent while FastICA employs fixed-point Newton iterations, we standardize resources via two defined regimes to ensure fairness:

- **Low Compute Baseline:** OT-ICA is allocated restarts $\min(4 \times d, 150)$ and 150/200 deflation/symmetric phase iterations. FastICA is allocated 50 random restarts and 1,000 maximum iterations per component.
- **High Compute Regime:** OT-ICA is allocated restarts $\min(15 \times d, 600)$ and 300/500 iterations. FastICA is allocated 50 restarts and 5,000 maximum iterations per component.

Algorithm 1 The Optimal Transport ICA (OT-ICA) Algorithm

Require: Observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, iterations K, learning rate η, batch size N_{batch}, dither noise σ_{dither}
Ensure: Estimated unmixing matrix $\mathbf{B}_{\text{final}} \in \mathbb{R}^{d \times d}$

Phase 1: Preprocessing & Target Generation
Center the data: $\mathbf{X} \leftarrow \mathbf{X} - \mathbb{E}[\mathbf{X}]$
Whiten the data: $\widetilde{\mathbf{X}} \leftarrow (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X}$
Compute analytical target $\mathbf{T} \in \mathbb{R}^{N_{\text{batch}}}$ via analytical Gaussian CDF integration
Phase 2: Initialization
Initialize $\mathbf{B} \in \mathbb{R}^{d \times d}$ randomly from $\mathcal{N}(0, 1)$
Apply symmetric decorrelation: $\mathbf{B} \leftarrow (\mathbf{B}\mathbf{B}^\top)^{-1/2}\mathbf{B}$
Phase 3: Stochastic Riemannian Optimization
for $k = 1$ to K **do**
    Sample stochastic mini-batch $\widetilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times N_{\text{batch}}}$ from $\widetilde{\mathbf{X}}$
    Project data onto candidates: $\mathbf{Y} \leftarrow \mathbf{B}\widetilde{\mathbf{X}}_{\text{batch}}$
    Apply Gaussian Dithering: $\mathbf{Y}_{\text{dither}} \leftarrow \mathbf{Y} + \mathcal{N}(0, \sigma^2_{\text{dither}})$
    Sort each row of $\mathbf{Y}_{\text{dither}}$ ascending to yield empirical quantiles $\mathbf{Y}_{\text{sorted}}$
    Compute Wasserstein objective: $\mathcal{L} = \frac{1}{N_{\text{batch}}} \sum_{i=1}^{d} ||\mathbf{Y}^{(i)}_{\text{sorted}} - \mathbf{T}||_2^2$
    Backpropagate to compute Euclidean gradient: $\mathbf{G} = \nabla_{\mathbf{B}}\mathcal{L}$
    Project to tangent space: $\nabla_{\mathcal{S}}\mathbf{B} = \mathbf{G} - \frac{1}{2}\big(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top\big)\mathbf{B}$
    Update matrix along tangent plane: $\mathbf{B}_{\text{step}} = \mathbf{B} + \eta\nabla_{\mathcal{S}}\mathbf{B}$
    Retract to Orthogonal Group $\mathcal{S}$: $\mathbf{B} \leftarrow (\mathbf{B}_{\text{step}}\mathbf{B}_{\text{step}}^\top)^{-1/2}\mathbf{B}_{\text{step}}$
end for
Phase 4: Finalization
$\mathbf{B}_{\text{final}} \leftarrow \mathbf{B}(\mathbf{X}\mathbf{X}^\top)^{-1/2}$
return $\mathbf{B}_{\text{final}}$

E LIMITATIONS OF PROXY CONTRAST FUNCTIONS

FastICA approximates negentropy using a non-quadratic contrast function $G(\cdot)$, such as the *logcosh* function. We identify two specific mathematical limitations of this approach.

E.1 THE ZERO NEGENTROPY CONDITION

The approximation isolates non-Gaussianity via the difference of expectations: $\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)]$, where $\nu \sim \mathcal{N}(0, 1)$. If a latent independent source possesses a non-Gaussian distribution such that its expected value under the contrast function perfectly equals that of a Gaussian, the approximated negentropy evaluates to zero. The objective function provides no gradient signal, and the algorithm fails to extract the source.

E.2 THE VANISHING CURVATURE CONDITION

FastICA utilizes a Newton (second-order) fixed-point iteration. Defining $g(x) = G'(x)$ and $g'(x) = G''(x)$, FastICA approximates the Hessian matrix dynamically. For a given component i, the diagonal entry scales according to the expectation $\mathbf{H}_{ii} \approx \mathbb{E}\big[s_i g(s_i) - g'(s_i)\big]$.

The Newton update requires applying the inverse of this Hessian ($\mathbf{H}^{-1}$). If a non-Gaussian source distribution causes the expectation in the denominator to evaluate to zero, the inverse becomes mathematically undefined. This analytical divide-by-zero error makes the update step intractable.

E.3 DISCRETE DISTRIBUTIONS: CONTRAST VS. OPTIMIZATION

The *Discrete Only* failure is mechanistically distinct from the continuous failure modes above. Table 2 compares the non-Gaussianity resolution of W_2^2 against the logcosh proxy negentropy $(\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)])^2$ across standard and highly non-Gaussian parameterizations.

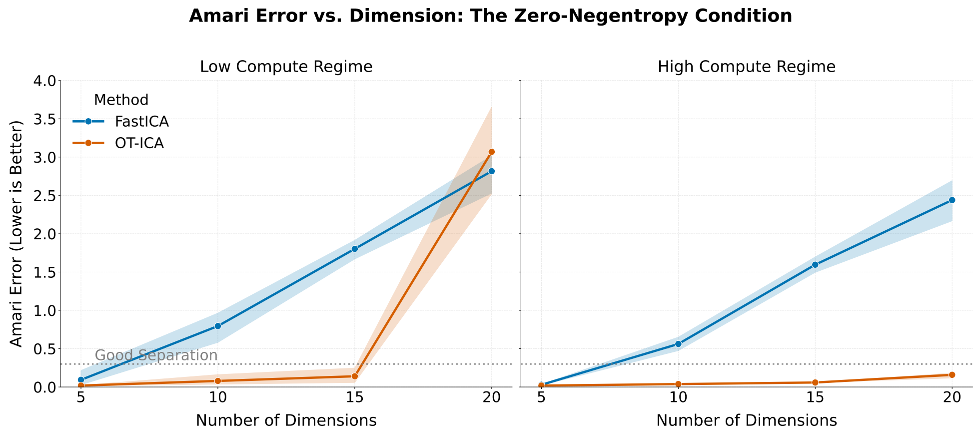

Figure 8: Amari error comparison for the engineered zero-negentropy distribution. While FastICA fails completely for $d > 5$, OT-ICA reliably maintains separation across higher dimensions.

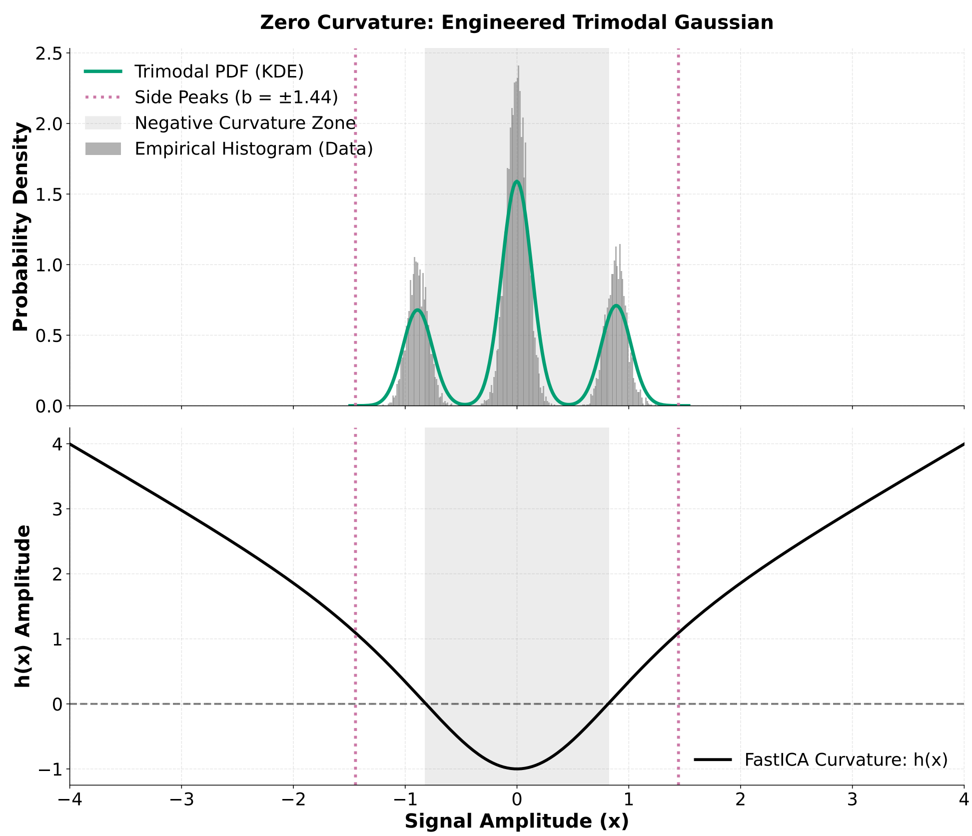

Figure 9: The Vanishing Curvature Counterexample. The central mass generates negative algorithmic curvature, while the side peaks generate positive algorithmic curvature. We solved for parameters such that these regions precisely cancel while maintaining unit variance.

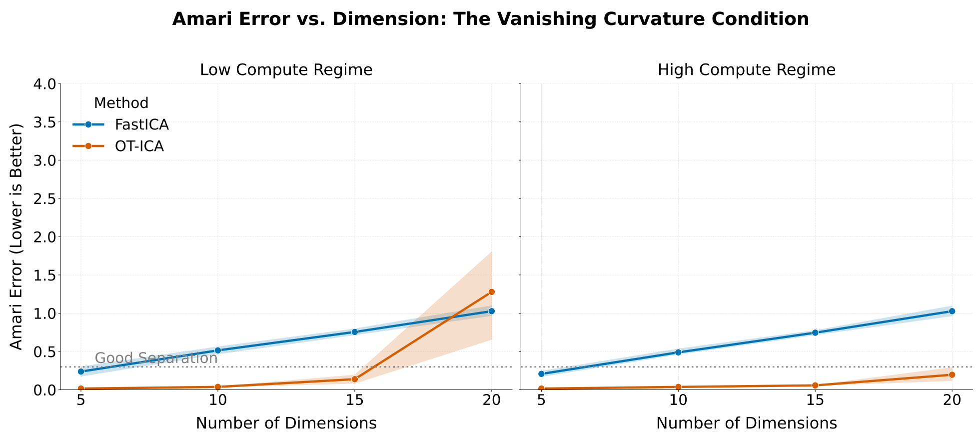

Figure 10: Amari error comparison for the zero-curvature distribution. FastICA fails entirely for $d \geq 10$. OT-ICA, utilizing the L-BFGS Quasi-Newton solver and the exact W_2^2 metric, bypasses the Hessian requirement and successfully maintains separation.

Distribution	$\mathbf{W_2^2}$	**Logcosh Negentropy**
Laplace (continuous baseline)	0.0398	0.001214
Binomial (standard, $n = 10$)	0.0344	0.000031
Poisson (standard, $\lambda = 3.0$)	0.0513	$\approx$0
Binomial (non-Gaussian, $n = 2$)	0.2022	0.000267
Poisson (non-Gaussian, $\lambda = 0.5$)	0.3384	0.000085

Table 2: W_2^2 provides $\gg$10$\times$ greater non-Gaussianity resolution than logcosh on count-based data. Standard Poisson ($\lambda = 3.0$) evaluates to near-zero under logcosh yet retains a clear W_2^2 signal. The failure to unmix discrete sources therefore cannot originate from an insufficient W_2^2 contrast; the issue is in the optimization landscape.

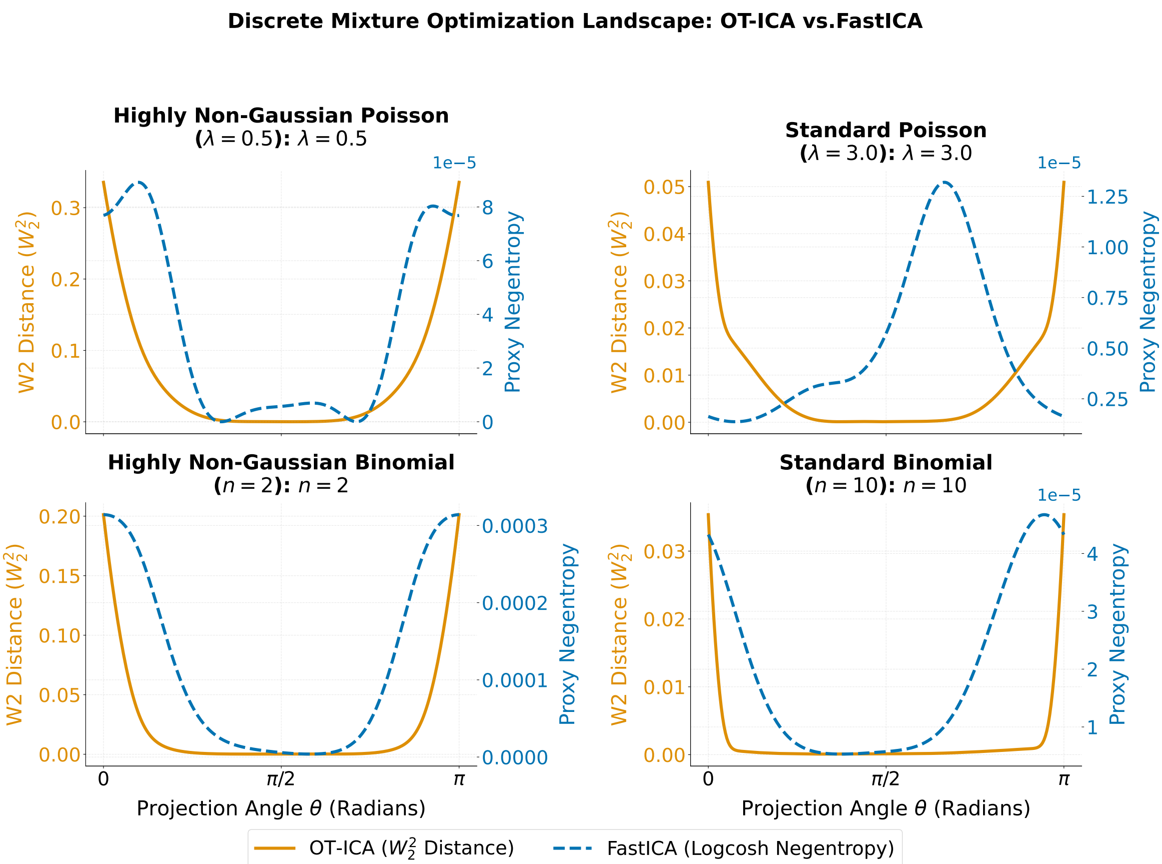

Figure 11: W_2^2 and logcosh optimization landscapes for two-source Poisson and Binomial mixtures across parameterizations. The W_2^2 landscape retains a non-zero signal throughout (confirming the contrast is present), but the step-function geometry of discrete CDFs creates flat plateaus where the gradient evaluates to near-zero, stalling gradient-based solvers. The logcosh landscape degenerates to near-zero for standard count-based parameterizations, indicating a fundamental contrast failure.

F COMPUTATIONAL SCALING AND DIMENSIONALITY LIMITS

We benchmarked OT-ICA against FastICA using a linear mixture of continuous Laplace sources across increasing dimensions with a fixed sample size of $N = 10,000$.

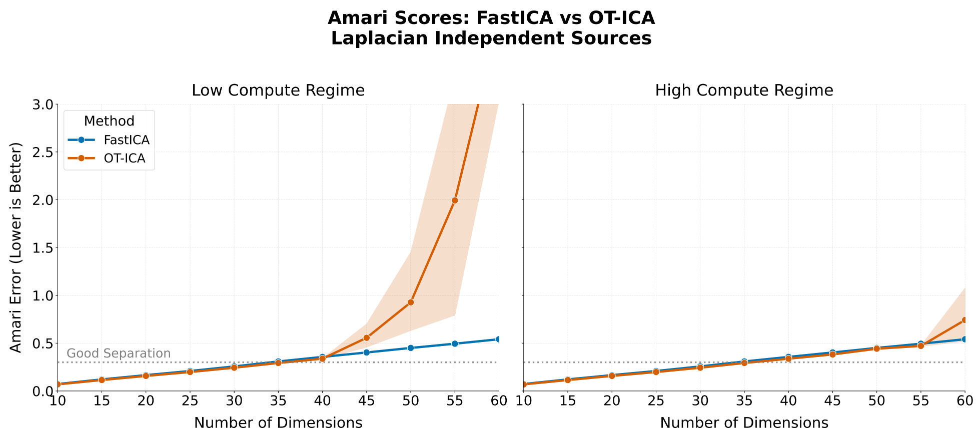

Figure 12: Amari error across dimensions for Laplacian sources at fixed $N = 10{,}000$ samples. Both OT-ICA and FastICA hit the same curse-of-dimensionality noise ceiling at $d = 40$—an artifact of finite-sample sparsity shared by all empirical ICA methods at this regime, not a deficiency specific to OT-ICA.

While OT-ICA maintains error rates comparable to FastICA ($E < 0.3$) in lower dimensions, both algorithms exhibit a loss in unmixing quality at $d = 40$. This limit is a manifestation of the curse of dimensionality. The expected error of the empirical Wasserstein distance scales as $\mathcal{O}(N^{-1/d})$. As d grows, a fixed finite sample fails to densely populate the state space, creating massive empty regions that swallow the gradient signal.

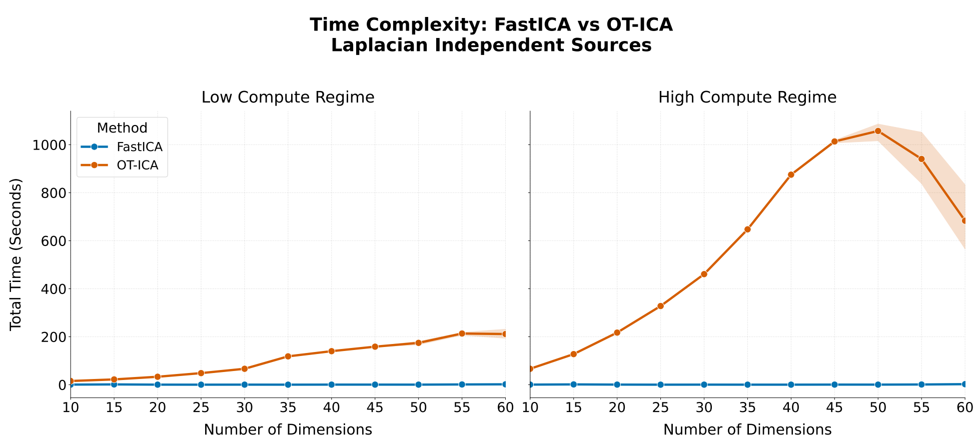

Figure 13: Total execution time (in seconds) for FastICA and OT-ICA across dimensions.

The execution time for OT-ICA scales more steeply than FastICA. FastICA evaluates to $\mathcal{O}(dN)$ per iteration, making it computationally superior for exclusively homogeneous, continuous mixtures that do not trigger its proxy blind spots. OT-ICA incorporates parallel sorting across multiple restarts ($\mathcal{O}(K \cdot dN \log N)$), trading execution speed for geometric robustness on complex topologies.